

Michael Cardei

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Education

University of Virginia | Ph.D. in Computer Science August 2024 – Expected May 2028
School of Engineering and Applied Science | **NSF GRFP Fellow**
Research Focus: Constrained Generative AI, Diffusion Large Language Models, and AI for Science
Advisor: Professor Ferdinando Fioretto

University of Florida | B.S. in Computer Science, *Cum Laude* June 2020 – May 2024
Herbert Wertheim College of Engineering | **GPA:** 3.92/4.00

Research Experience

Graduate Research Assistant August 2024 – Present
University of Virginia, RAISE Lab, Advised by Professor Ferdinando Fioretto

- Researching the topics of Generative AI for Scientific and Engineering Applications in addition to Responsible Generative AI.
- Developing foundational methodologies to enable the integration of **constrained optimization within Discrete Diffusion Models**.

Graduate Student AI Research Intern May 2026 – Present
Lawrence Livermore National Lab, Livermore, California (On-site)

- Research **reinforcement learning-based** post-training methods to enhance large language models for **scientific discovery** and domain-specific reasoning.

Staff Machine Learning Research Scientist Intern May 2025 – August 2025
Visa, Foster City, California (On-site)

- Developed diffusion-based generative modeling methods for constrained synthetic data generation in financial applications.

Research Intern August 2023 – May 2024
University of Florida, Adaptive Learning and Optimization Lab, Advised by Dr. My Thai

- Investigated **privacy vulnerabilities** and explored implementation strategies within Federated Learning for **Large Language Models**.
- Examining neuron-based explainable AI methods for network intrusion anomaly detection mechanism analysis.

AI Research Intern (Robotics Institute Summer Scholars (RISS)) June 2023 – August 2023
Carnegie Mellon University Robotics Institution, ILIM Lab, Advised by Dr. Srinivasa Narasimhan

- Researched methods for context-driven road work-zone detection and localization for autonomous vehicles.
- Leveraged advanced **Computer Vision, Deep Learning, and NLP** techniques, including detection, instance segmentation, scene text recognition, and transfer learning.

Research Intern August 2022 – June 2023
Wake Forest University, Advised by Dr. Umit Topaloglu

- Researched novel methods for **bias mitigation and fairness** in medical deep learning applications
- Implemented, optimized, and tested deep learning algorithms while also performing feature engineering, model creation, and model evaluation
- Used multiple Machine Learning frameworks such as TensorFlow, PyTorch, and Keras for the creation and implementation of Deep Neural Networks

Research Intern (REU) May 2022 – August 2022
Wake Forest University School of Medicine, Advised by Dr. Umit Topaloglu

- Researched novel approaches for **Privacy Preserved Machine Learning** based upon data frequency domain transformations
- Created and tested multiple adversarial attacks along with implementing the privacy methods in a **Federated learning** environment. Utilized TensorFlow Federated and TensorFlow Privacy along with other machine learning libraries.

Publications

In computer science, peer-reviewed conference proceedings are the primary venue for disseminating research.

Conference Papers

- [Constrained Discrete Diffusion](#) 
M. Cardei, J. K. Christopher, T. Hartvigsen, B. Kailkhura, and F. Fioretto.
Advances in Neural Information Processing Systems (NeurIPS) 2025.
- [Constrained Tabular Diffusion for Finance](#) 
M. Cardei, J. M. Munoz, O. Barrera, S. K. Chandrabhas, P. Saha.
ACM International Conference on AI in Finance (ICAIF) 2025
- [Neuro-Symbolic Generative Diffusion Models for Physically Grounded, Robust, and Safe Generation](#) 
J. Christopher, M. Cardei, J. Liang, F. Fioretto.
2nd International Conference on Neuro-Symbolic Systems (NeuS) 2025
DARPA Disruptive Idea Award (Best Paper).
- [Speculative Diffusion Decoding: Accelerating Language Generation through Diffusion](#) 
J. Christopher, B. Bartoldson, T. Ben-Nun, M. Cardei, B. Kailkhura, F. Fioretto.
North American Chapter of the Association for Computational Linguistics (NAACL) 2025 **Oral.**
- [ROADWork Dataset: Learning to Recognize, Observe, Analyze and Drive Through Work Zones](#) 
A. Ghosh, R. Tamburo, S. Zheng, J. Alvarez-Padilla, H. Zhu, M. Cardei, N. Dunn, C. Mertz, S. Narasimhan.
International Conference on Computer Vision (ICCV) 2025.
- [Improving Equity in Deep Learning Medical Applications with the Gerchberg-Saxton Algorithm](#) 
S. Ay, M. Cardei, A. Meyer, W. Zhang and U. Topaloglu. *IEEE 11th International Conference on Healthcare Informatics (ICHI) 2023.*
- [Automatic Detection and Localization of Roadwork](#) 
S. Narasimhan, R. Tamburo, C. Mertz, D. Reddy, K. Vuong, A. Ghosh, S. Srivastava, N. Bolor, T. Ma, M. Cardei, N. Dunn, H. Zhu, Automatic Detection and Localization of Roadwork
Mobility21, Carnegie Mellon University, 2023.

Journal Articles

- [Improving Equity in Deep Learning Medical Applications with the Gerchberg-Saxton Algorithm](#) 
S. Ay, M. Cardei, A. Meyer, W. Zhang, and U. Topaloglu.
Journal of Healthcare Informatics Research (2024).

Workshop Papers

- [Constrained Molecular Generation with Discrete Diffusion for Drug Discovery](#) 
M. Cardei, J. K. Christopher, T. Hartvigsen, B. Kailkhura, and F. Fioretto.
AI Virtual Cells and Instruments: A New Era in Drug Discovery and Development Workshop (NeurIPS 2025).
Best Student Paper Award.

- [Compliant Tabular Diffusion for Finance](#) 
M. Cardei, J. M. Munoz, O. Barrera, S. K. Chandrahas, P. Saha.
Generative AI in Finance Workshop (NeurIPS 2025).

Preprints / Under Review

- [Search-Augmented Masked Diffusion Models for Constrained Generation](#) 
M. Cardei*, H. Binh-Ta*, A. Velasquez, and F. Fioretto.
- [Simple Self-Conditioning Adaptation for Masked Diffusion Models](#) 
M. Cardei*, H. Binh-Ta*, and F. Fioretto.
- [Constrained Code Generation with Discrete Diffusion](#) 
M. Cardei*, L. Shao*, Z. Xie, W. Wang, and F. Fioretto.

Achievements and Awards

- National Science Foundation Graduate Research Fellowship Program (NSF GRFP), 2025
- NeurIPS Scholar Award, 2025
- Best Student Paper Award, NeurIPS AI4D3 Workshop, 2025, for “Constrained Molecular Generation with Discrete Diffusion for Drug Discovery”
- DARPA Disruptive Idea Award / Best Paper Award, NeuS, 2025, for “Neuro-Symbolic Generative Diffusion Models for Physically Grounded, Robust, and Safe Generation”
- Best Paper Award, UVA LLM Workshop, 2024, for “Speculative Diffusion Decoding”
- University of Virginia Computer Science Scholar, 2024–2025
- Carnegie Mellon University Robotics Institute Summer Scholar, 2023
- 2nd Place Research Presentation, Wake Forest Summer Symposium, Cancer, Imaging, and Informatics Category
- 3rd Place Finalist, University of Florida Artificial Intelligence Hackathon
- Wake Forest University BME and Informatics Summer Research Scholar

Teaching Assistant

CS 6501 Constrained-Aware Gen AI University of Virginia Spring 2026
CS4710 Artificial Intelligence University of Virginia Fall 2025

Academic Service

Reviewer

Advances in Neural Information Processing Systems (NeurIPS), 2026
 ACM International Conference on AI in Finance (ICAIF), 2025

Relevant Courses

University of Virginia (Ph.D.)

Graph Machine Learning, Natural Language Processing, Machine Learning in Systems Security, Machine Learning for Image Analysis, Machine Learning for Software Reliability

University of Florida (B.S.)

Trustworthy Machine Learning (Graduate Course), Applied Machine Learning, Natural Language Processing, Introduction to Multi-Modal Machine Learning, Programming Language Concepts, Engineering Statistics, Operating Systems, Data Structures and Algorithms

Skills

- Core Competencies in AI: Generative AI, Large Language Models, Diffusion Language Models, Diffusion Models, Privacy, Bias Reduction, Deep Learning, Computer Vision, Federated Learning, Constrained Optimization
- Languages: Python, C++, Java, R, SQL
- Tools/Frameworks: Pytorch, TensorFlow, Keras, MMDetection, Mask2Former, Scikit-Learn, TensorFlow Federated, TensorFlow Privacy, MongoDB, GitHub, Huggingface Transformers